2.1.2 Water and its importance in plants and animals

		Learning outcomes	Additional guidance
		Learners should be able to demonstrate and apply their knowledge and understanding of	
(a)		the properties of water	To include the polar nature of the water molecule, hydrogen bonding and the role of water as a solvent.
(b)	**(i)**	the importance of water as a major constituent of cytoplasm, intracellular and extracellular fluids, and as the essential transport medium in plants and animals	To include the transpiration stream, cell sap and the maintenance of turgor in plants, and plasma, serum, tissue fluid, lymph and urine in mammals. *M0.1, M0.2, M0.3, M1.1, M1.2, M1.6, M3.1* HSW5, HSW6
	(ii)	analysis of secondary data on the composition of mammalian body fluids and plant extracts to illustrate the role of water as a solvent	To include solutes (sugars, proteins), electrolytes (hydrogen ions, H^+, potassium ions, K^+, sodium ions, Na^+, chloride ions, Cl^-, hydrogencarbonate ions, HCO_3^-, magnesium ions, Mg^{2+}).
(c)	**(i)**	how sugar and protein molecules can be detected and measured in body fluids and plant extracts	To include the use of reagent test strips and biosensors to detect and measure the concentration of sugars and proteins. *M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M3.1, M3.2* **PAG5, PAG9** HSW3, HSW4, HSW5, HSW6
	(ii)	the methodology and interpretation of the results of the Biuret test, Benedict's test and colorimetry	
(d)		the importance of hydrolysis and condensation of biological molecules in cell metabolism	To include the concept of monomers and polymers in a range of biological molecules.
(e)		the structure of the ring form of α-glucose as an example of a simple monosaccharide, and lactose as a disaccharide	To include the concept of organic molecules as generally containing carbon atoms and a number of additional elements.
(f)	**(i)**	the formation of polysaccharides by condensation	To include glycogen and starch (amylose and amylopectin) **AND** the formation of 1,4- and 1,6-glycosidic bonds and reference to the significance of branching on solubility.
	(ii)	a test for the identification and measurement of starch	To include the qualitative test for starch using iodine and colorimetry. **PAG5, PAG9**

(g)	osmosis, in terms of the movement of water down a water potential gradient	To include the effect of solutes and electrolytes on the water potential of plant cells and animal cells and on solutions within organisms e.g. body fluids, plant sap. *M0.3, M1.1, M1.2, M1.3, M3.1, M3.2, M3.4*
(h)	practical investigation(s) into factors affecting osmosis in plant and animal cells.	*M0.3, M1.1, M1.2, M1.3, M3.1, M3.2, M3.4* **PAG8** HSW3, HSW4, HSW5, HSW6